

Program : Diploma in Artificial Intelligence & Machine Learning	
Course Code : 4348	Course Title: Object Oriented Programming in Java Lab
Semester : 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- Students should be able to understand the concepts of object oriented programming and implement simple JAVA applications. .

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic Mathematics		Mathematics	1, 2
Programming concepts.		Introduction to IT Systems Lab	2
Programming concepts.		Programming in C	2

Course Outcomes :

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Implement simple Java programs	11	Applying
CO2	Implement different types of inheritances	10	Applying
CO3	Implement exception handling and package creation	9	Applying
CO4	Implement GUI Applications	12	Applying
	Lab Exam	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3	3	3	3	3	3	3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of the Experiment	Duration (Hours)	Cognitive Level
CO1	Implement simple Java programs		
M1.01	Develop java programs using class and objects	2	Applying
M1.02	Implement java programs that handle array of objects	2	Applying
M1.03	Implement java programs passing and returning objects as arguments.	2	Applying
M1.04	Implement programs based on I/O operations using I/O classes.	2	Applying
M1.05	Demonstrate constructor and method overloading.	3	Applying
CO2	Implement different types of inheritances		
M2.01	Implement programs using single inheritance	2	Applying
M2.02	Implement programs using multilevel and hierarchical inheritance	3	Applying
M2.03	Implement method overriding	2	Applying
M2.04	Implement multiple inheritance using interfaces	3	Applying
	Lab Exam – I	1.5	
CO3	Implement exception handling and package creation		
M3.01	Implement exception handling using try-catch	2	Applying

M3.02	Implement exception handling using multiple catch	2	Applying
M3.03	Implement exception handling using nested try	1	Applying
M3.04	Implement throwing an Exception	1	Applying
M3.05	Implement user defined packages	3	Applying
CO4	GUI Programming		
M4.01	Implement simple GUI programs with java applets	3	Applying
M4.02	Implement event handling in java applet GUI.	3	Applying
M4.03	Open Ended Experiments**	6	Applying
	Lab Exam – II	1.5	

**** - Suggested Open Ended Experiments**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open ended experiments as a group of 2-3. There is no duplication in experiments between groups.)

1. **Bank account management:** Create a class for the bank account, which would contain methods for depositing, withdrawing, and transferring money. There are different types of accounts such as savings account, current account, and fixed deposit account. Each of these accounts has common features such as account number, account holder name, and account balance. However, they also have unique features such as interest rate for savings account, overdraft facility for current account, and maturity period for fixed deposit account. Also create a class for the customer, which would contain information such as the name, address, and account number.
2. **Vehicle Management:** Consider a scenario where there are different types of vehicles such as cars, trucks, and motorcycles. Each of these vehicles has common properties such as a manufacturer, model, and year of production. However, they also have unique properties such as the number of doors for cars, the maximum load capacity for trucks, and the number of wheels for motorcycles. Implement the scenario.
3. Create a simple Java GUI application

Text / Reference

T/R	Book Title/Author
T1	Balaguruswamy E: Programming with Java, 6th edition.
T2	Herbert Schildt, “Java the complete reference”, McGraw Hill, Osborne, 7th Edition, 2011
R1	Herbert Schildt and Dale Skrien, ”Java Fundamentals – A comprehensive Introduction”, McGraw Hill, 1st Edition, 2013.
R2	T.Budd, “Understanding Object- Oriented Programming with Java”, Pearson Education, Updated Edition (New Java 2 Coverage), 1999

Online Resources

Sl.No	Website Link
1	http://java.sun.com 2. http://www.oracle.com/technetwork/java/index.html)
2	http://www.oracle.com/technetwork/java/javase/overview/index.html 5.
3	http://download.oracle.com/javase/7/docs/api/index.html